

STAT 516 - Homework 4

Instructions: Please submit your homework as a single PDF on Gradescope. After uploading, please make sure each page is assigned to the correct question before submitting. Show your work clearly.

1. **(9 points)** A discrete random variable X has probability mass function

$$P(X = x) = \begin{cases} p, & x = 0, \\ 2p, & x = 2, \\ 3p, & x = 4, \\ 1 - 6p, & x = 6. \end{cases}$$

You are given that $E[X] = 3.6$. Determine p .

2. **(9 points)** A discrete random variable X takes values 1, 2, 4, 5 with probabilities

$$P(X = 1) = 0.10, \quad P(X = 2) = 0.25, \quad P(X = 4) = 0.40, \quad P(X = 5) = 0.25.$$

Calculate $E[X]$ and $\text{Var}(X)$.

3. **(10 points)** Let X be a continuous random variable with density function

$$f(x) = \begin{cases} cx^2, & 0 < x < 2, \\ 0, & \text{otherwise.} \end{cases}$$

Find c , and then calculate $E[X]$.

4. **(9 points)** Let X be a continuous random variable with density function

$$f(x) = \begin{cases} \frac{2|x|}{13}, & -2 \leq x \leq 3, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate $E[X]$.

5. **(9 points)** Let X be a continuous random variable with density function

$$f(x) = \begin{cases} \frac{3}{26}x^2, & 1 < x < 3, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate $E\left[\frac{1}{X}\right]$.

6. **(9 points)** Let X be a continuous random variable with density function

$$f(x) = \begin{cases} \lambda^2 x + \frac{3}{2}\lambda^3 x^2, & 0 < x < \frac{1}{\lambda}, \\ 0, & \text{otherwise,} \end{cases}$$

where $\lambda > 0$. Verify that this is a valid density function.

7. **(10 points)** Let X be a continuous random variable with cumulative distribution function

$$F(x) = \begin{cases} 0, & x < 1, \\ \frac{(x-1)^2}{9}, & 1 \leq x < 4, \\ 1, & x \geq 4. \end{cases}$$

Calculate $P(X < 3)$ and find the density function $f(x)$.

8. **(10 points)** Let X be a continuous random variable with density function

$$f(x) = \begin{cases} \frac{1}{3}e^{-x/3}, & x > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate $E[X]$.

9. **(9 points)** You are given

$$E[X] = 6, \quad \text{Var}(X) = 16.$$

Calculate

$$E[(2X - 3)^2].$$

10. **(9 points)** Let X be a random variable such that

$$E[X] = 3, \quad E[X^3] = 36, \quad E[(X - 3)^3] = 0.$$

Calculate $\text{Var}(X)$.

11. **(7 points)** The annual repair cost X for a machine has mean 500 dollars and variance 900. After a price increase, the new repair cost is

$$Y = 1.15X + 20.$$

Calculate $\text{Var}(Y)$.

1. (9 points) A discrete random variable X has probability mass function

$$P(X = x) = \begin{cases} p, & x = 0, \\ 2p, & x = 2, \\ 3p, & x = 4, \\ 1 - 6p, & x = 6. \end{cases}$$

You are given that $E[X] = 3.6$. Determine p .

$$E[X] = \sum xP(X=x)$$

$$E[X] = 0p + (2p \cdot 2) + (3p \cdot 4) + ((1-6p) \cdot 6) = 4p + 12p + (6-36p) = 3.6$$

$$6-20p = 3.6$$

$$p = 0.12$$

2. (9 points) A discrete random variable X takes values 1, 2, 4, 5 with probabilities

$$P(X = 1) = 0.10, \quad P(X = 2) = 0.25, \quad P(X = 4) = 0.40, \quad P(X = 5) = 0.25.$$

Calculate $E[X]$ and $\text{Var}(X)$.

$$E[X] = (1 \cdot 0.10) + (2 \cdot 0.25) + (4 \cdot 0.40) + (5 \cdot 0.25)$$

$$\text{Var}(X) = E[X^2] - E[X]^2 = 13.75 - (3.45)^2$$

$$E[X] = 3.45$$

$$\text{Var}(X) = 1.8475$$

$$E[X^2] = (1^2 \cdot 0.10) + (2^2 \cdot 0.25) + (4^2 \cdot 0.40) + (5^2 \cdot 0.25)$$

$$E[X^2] = 13.75$$

3. (10 points) Let X be a continuous random variable with density function

$$f(x) = \begin{cases} cx^2, & 0 < x < 2, \\ 0, & \text{otherwise.} \end{cases}$$

Find c , and then calculate $E[X]$.

$$f(x) = \int_0^2 cx^2 dx = c \int_0^2 x^2 dx = c \left[\frac{x^3}{3} \right]_0^2 = c \left(\frac{2^3}{3} - \frac{0^3}{3} \right) = c \left(\frac{8}{3} \right) = 1$$

$$c = 3/8$$

$$E[X] = \int x f(x) dx$$

$$E[X] = \int_0^2 x \left(\frac{3}{8} \right) x^2 dx = \frac{3}{8} \int_0^2 x^3 dx = \frac{3}{8} \left[\frac{x^4}{4} \right]_0^2 = \frac{3}{8} \left(\frac{2^4}{4} \right)$$

$$E[X] = 3/2$$

4. (9 points) Let X be a continuous random variable with density function

$$f(x) = \begin{cases} \frac{2|x|}{13}, & -2 \leq x \leq 3, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate $E[X]$.

$$E[X] = \int x f(x) dx$$

$$E[X] = \int_{-2}^3 x \left(\frac{2|x|}{13} \right) dx$$

→ Absolute value split at 0:

$$|x| = -x \text{ for } x < 0$$

$$|x| = x \text{ for } x > 0$$

$$= \int_{-2}^0 x \left(\frac{2|-x|}{13} \right) dx + \int_0^3 x \left(\frac{2|x|}{13} \right) dx = \int_{-2}^0 \frac{-2x^2}{13} dx + \int_0^3 \frac{2x^2}{13} dx = \frac{-2}{13} \int_{-2}^0 x^2 dx + \frac{2}{13} \int_0^3 x^2 dx$$

$$= \frac{-2}{13} \left[\frac{x^3}{3} \right]_{-2}^0 + \frac{2}{13} \left[\frac{x^3}{3} \right]_0^3 = \frac{-2}{13} \left(\frac{8}{3} \right) + \frac{2}{13} (9)$$

$$E[X] = 38/39$$

5. (9 points) Let X be a continuous random variable with density function

$$f(x) = \begin{cases} \frac{3}{26} x^2, & 1 < x < 3, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate $E\left[\frac{1}{X}\right]$.

$$E[X] = \int x f(x) dx, \text{ then } E\left[\frac{1}{X}\right] = \int \frac{1}{x} f(x) dx$$

$$E\left[\frac{1}{X}\right] = \int_1^3 \frac{1}{x} \left(\frac{3}{26} \right) x^2 dx = \int_1^3 \frac{3x}{26} dx = \frac{3}{26} \int_1^3 x dx$$

$$= \frac{3}{26} \left[\frac{x^2}{2} \right]_1^3 = \frac{3}{26} \left(\frac{9}{2} - \frac{1}{2} \right)$$

$$E\left[\frac{1}{X}\right] = 6/13$$

6. (9 points) Let X be a continuous random variable with density function

$$f(x) = \begin{cases} \lambda^2 x + \frac{3}{2} \lambda^3 x^2, & 0 < x < \frac{1}{\lambda}, \\ 0, & \text{otherwise,} \end{cases}$$

where $\lambda > 0$. Verify that this is a valid density function.

$f(x) \geq 0$ since $\lambda > 0, x > 0$.

$$\begin{aligned} \int_0^{1/\lambda} (\lambda^2 x + \frac{3}{2} \lambda^3 x^2) dx &= \int_0^{1/\lambda} \lambda^2 x dx + \int_0^{1/\lambda} \frac{3}{2} \lambda^3 x^2 dx = \lambda^2 \left[\frac{x^2}{2} \right]_0^{1/\lambda} + \frac{3}{2} \lambda^3 \left[\frac{x^3}{3} \right]_0^{1/\lambda} \\ &= \lambda^2 \left(\frac{(1/\lambda)^2}{2} \right) + \frac{3}{2} \lambda^3 \left(\frac{(1/\lambda)^3}{3} \right) = \lambda^2 \left(\frac{1}{2\lambda^2} \right) + \frac{3}{2} \lambda^3 \cdot \frac{1}{3\lambda^3} \\ &= \frac{1}{2} + \frac{1}{2} = 1 \quad \checkmark \end{aligned}$$

$$\int_{-\infty}^{\infty} f(x) dx = 1, \text{ therefore this } f(x) \text{ is a valid PDF.}$$

7. (10 points) Let X be a continuous random variable with cumulative distribution function

$$F(x) = \begin{cases} 0, & x < 1, & (-\infty, 1) \\ \frac{(x-1)^2}{9}, & 1 \leq x < 4, & [1, 4) \\ 1, & x \geq 4. & [4, \infty) \end{cases}$$

Calculate $P(X < 3)$ and find the density function $f(x)$.

$$\begin{aligned} F(x) &= P(X \leq x) \\ P(X < 3) &= P(X \leq 3) = F(3) = \frac{(3-1)^2}{9} = \frac{2^2}{9} \\ P(X \leq 3) &= 4/9 \end{aligned}$$

$$f(x) = F'(x)$$

$$\begin{aligned} \textcircled{1} \quad x < 1: \quad F(x) &= 0, \quad F'(x) = 0 \\ \textcircled{2} \quad 1 \leq x < 4: \quad F(x) &= \frac{(x-1)^2}{9}, \quad F'(x) = \frac{2(x-1)}{9} \\ \textcircled{3} \quad x \geq 4: \quad F(x) &= 1, \quad F'(x) = 0 \end{aligned}$$

For density, use open interval

$$f(x) = \begin{cases} \frac{2(x-1)}{9}, & 1 < x < 4 \\ 0, & \text{otherwise} \end{cases}$$

8. (10 points) Let X be a continuous random variable with density function

$$f(x) = \begin{cases} \frac{1}{3}e^{-x/3}, & x > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate $E[X]$.

$$E[X] = \int_{-\infty}^{\infty} x f(x) dx$$

Exponential distribution
with rate: $\lambda = 1/3$

$$\int_0^{\infty} x e^{-ax} dx = 1/a^2$$

$a = 1/3$

$$E[X] = \int_0^{\infty} x \left(\frac{1}{3} e^{-x/3} \right) dx = \frac{1}{3} \int_0^{\infty} x e^{-x/3} dx = \frac{1}{3} \left(\frac{1}{(1/3)^2} \right) = \frac{1}{3} \cdot 9$$

$$E[X] = 3$$

9. (9 points) You are given

$$E[X] = 6, \quad \text{Var}(X) = 16.$$

Calculate

$$E[(2X - 3)^2].$$

$$\text{Var}(X) = E[X^2] - (E[X])^2 = 16$$

$$E[X^2] - (6)^2 = 16$$

$$E[X^2] = 52$$

$$E[(2X - 3)^2] = E[4X^2 - 12X + 9]$$

$$= 4E[X^2] - 12E[X] + 9 = 4(52) - 12(6) + 9$$

$$E[(2X - 3)^2] = 145$$

10. (9 points) Let X be a random variable such that

$$E[X] = 3, \quad E[X^3] = 36, \quad E[(X - 3)^3] = 0.$$

Calculate $\text{Var}(X)$.

$$\begin{aligned} E[(X-3)^3] &= E[X^3 - 9X^2 + 27X - 27] = 0 \\ &= E[X^3] - 9E[X^2] + 27E[X] - 27 = 0 \\ &= 36 - 9E[X^2] + 27(3) - 27 = 0 \\ &= 90 - 9E[X^2] = 0 \\ E[X^2] &= 10 \end{aligned}$$

$$\text{Var}(X) = E[X^2] - (E[X])^2 = 10 - (3)^2 = 10 - 9$$

$$\text{Var}(X) = 1$$

11. (7 points) The annual repair cost X for a machine has mean 500 dollars and variance 900. After a price increase, the new repair cost is

$$Y = 1.15X + 20.$$

Calculate $\text{Var}(Y)$.

$$E[X] = 500$$

$$\text{Var}(X) = 900$$

$$\text{Variance Rule: } \text{Var}(aX+b) = a^2 \text{Var}(X)$$



$$\begin{aligned} a &= 1.15 \\ b &= 20 \end{aligned}$$

$$\text{Var}(Y) = \text{Var}(1.15X + 20) = 1.15^2 \text{Var}(X) = 1.15^2(900)$$

$$\text{Var}(Y) = 1190.25$$